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Embedding Geometric Heuristics for Improving Retrieval-Augmented Generation

Budget-Aware Search in the Embedding Space Guided by an LLM Judge

Research Proposal

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Abstract

In Retrieval-Augmented Generation (RAG), the quality of the answer is bounded by the quality of retrieval: if the relevant document is not retrieved, the generator cannot use it. Dense retrievers rank documents by embedding similarity to the query. They are cheap, but in specialised domains they miss the target document on a non-trivial fraction of queries. LLM judges score relevance much more accurately, but every call is expensive, and their scores are coarse: many documents receive the same top score.

This research treats retrieval with an LLM judge as a *budget-constrained sequential search problem*. Embedding distance is a cheap surrogate, the judge is an expensive oracle, and the policy chooses which document to judge next so as to reach the target in as few calls as possible. Our preliminary study used 196 PubMedQA questions over a shared pool of 5,035 documents drawn from a full-PubMed dense index, with all ≈1M question–document pairs scored by Gemini 2.5 Flash-Lite. It showed that (i) the judge's score is a reliable but coarse proxy for proximity to the target: its correlation with distance is negative in every query and ROC-AUC is 0.997, but a median of 18 documents per query share the top score; (ii) queries are geometric outliers with respect to the documents; and (iii) the centroid of the judge-approved documents is 44% closer to the target than the query is, yet static use of this centroid improves top-1 accuracy by at most one percentage point over the dense retriever (70.4%→71.4%), and naive online steering towards it increases cost.

Building on these findings, we will develop and evaluate (a) judge-guided search policies that invoke the expensive search only when the retriever is likely to have failed, use repulsive feedback from relevant-but-non-target documents, and localise the target by trilateration from judge scores; (b) finer-grained judging that breaks score ties; and (c) a characterisation and early prediction of retrieval failures. Policies will be compared by their cost–accuracy curves with dense retrieval, re-ranking, pseudo-relevance feedback and HyDE, on biomedical and general-domain benchmarks.

Keywords: retrieval-augmented generation, dense retrieval, embedding geometry, LLM-as-a-judge, budget-constrained search, biomedical question answering.

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1. Introduction

Large language models (LLMs) store knowledge in their parameters. That knowledge is fixed at training time and incomplete in specialised domains. Retrieval-Augmented Generation (RAG) addresses this by retrieving documents from an external corpus at inference time and conditioning the LLM on them (Lewis et al., 2020; Gao et al., 2023b). A RAG pipeline is only as good as its retriever: if the document that answers the question is not retrieved, the generator either cannot answer or answers from irrelevant context, which can actively mislead it (Shi et al., 2023). In biomedicine, where RAG is increasingly used for clinical and scientific question answering (Xiong et al., 2024), such failures can lead to confidently wrong answers.

Two kinds of retrieval signal are available today, and they behave very differently.

- **Embedding similarity.** Dense bi-encoders embed queries and documents in a shared vector space and rank documents by cosine similarity to the query (Karpukhin et al., 2020; Reimers and Gurevych, 2019). With an approximate nearest-neighbour index the ranking is almost free (Malkov and Yashunin, 2020). It is also systematically imperfect: on our challenging biomedical benchmark the top-ranked document is the correct one for only 70% of the questions (Chapter 5).
- **LLM judgement.** An LLM prompted to rate the relevance of a document to a question separates relevant from irrelevant documents almost perfectly (Thomas et al., 2024; Faggioli et al., 2023). However, each judgement is an LLM call, and its integer scale produces large groups of documents with the same top score (Zhuang et al., 2024).

The obvious combination, retrieving the top-k documents and re-ranking them with the LLM, uses the judge as a passive scorer of a fixed list. That approach pays for k calls whether or not the retriever was already right. It cannot reach a target that lies outside the initial list, and it inherits the judge's ties. This research starts from a different view, developed during our supervision meetings: *the judge's scores are feedback that can steer a search through the embedding space.* Each judged document tells us something about where the target lies. This resembles the word game Semantle, where a player guesses a hidden word from the semantic similarity of previous guesses, and trilateration, where a position is recovered from distances to known anchors. The

research question then becomes algorithmic: *given a cheap but imperfect geometric surrogate and an accurate but expensive judge, which document should be judged next, so that the target is found with as few judge calls as possible?*

What we have learned so far. Over the past two years we built the experimental infrastructure for this question: a full-PubMed dense index, a cloud and cluster pipeline that scored about one million question–document pairs with an LLM judge, and an offline simulator that replays any search policy against these scores at no additional cost. With this infrastructure we established the following (Chapter 5):

1. The judge's score is a reliable but coarse signal of proximity to the target document. The correlation with distance is negative for every query and ROC-AUC is 0.997, yet a median of 18 documents per query share the top score.
2. Queries are geometric outliers relative to documents. The neighbourhood of a query is almost empty, while the neighbourhood of its target document is populated.
3. The centroid of the documents the judge approves is 44% closer to the target than the query is. Nevertheless, *static* heuristics that use it improve top-1 accuracy by at most one percentage point, and *naive* online steering towards it increases the number of judge calls.

Together, these results show that the geometric signal is real, that it is not exploited by simply mixing distances, and that the open problem is a well-posed search problem.

Research questions. The proposed research addresses five questions, detailed in Chapter 4. How reliable is the LLM judge as a proximity signal, and can its resolution be improved (RQ1)? What geometric properties distinguish queries on which dense retrieval fails (RQ2)? Can judge-guided search policies find the target with fewer calls than the retriever's order (RQ3)? Can failure be detected early enough to invoke the search only when needed (RQ4)? Do the findings generalise across embedders, judges and datasets (RQ5)?

Structure of the proposal. Chapter 2 reviews the relevant literature. Chapter 3 formalises the problem. Chapter 4 states the objectives, research questions, hypotheses and expected contributions. Chapter 5 reports the preliminary results, Chapter 6 presents the methodology and work plan, and Chapter 7 gives the time schedule.

2. Scientific Background and Related Work

2.1 Retrieval-Augmented Generation

A RAG system has three stages (Lewis et al., 2020; Gao et al., 2023b). An *indexer* embeds the corpus offline. A *retriever* returns the documents most similar to the query. A *generator* LLM produces the answer conditioned on them. "Modular" RAG architectures add pre-retrieval steps (query rewriting) and post-retrieval steps (re-ranking, filtering, compression) around the retriever (Gao et al., 2023b). These additions exist because the generator is sensitive to what it receives. Irrelevant passages distract it (Shi et al., 2023), and relevant information placed in the middle of a long context is often ignored (Liu et al., 2024). Black-box generators can be improved by tuning the retriever alone (Shi et al., 2024), which confirms that retrieval is a lever on end-to-end quality. In medicine, the MIRAGE benchmark and the MedRAG toolkit showed that the choice of corpus, retriever and number of retrieved snippets changes answer accuracy substantially (Xiong et al., 2024). BERGEN provides a comparable benchmarking library for general-domain RAG (Rau et al., 2024). RAG outputs are evaluated by humans, by model-based judges (Zheng et al., 2023; Es et al., 2024), or component by component. We follow the component-level view and measure retrieval directly, because retrieval errors bound everything downstream.

2.2 Dense Retrieval and Nearest-Neighbour Search

Dense retrievers encode queries and documents independently with a bi-encoder and score them by inner product or cosine similarity (Karpukhin et al., 2020; Reimers and Gurevych, 2019). Document vectors are pre-computed and stored in a vector database. At query time, a graph-based index such as HNSW (Malkov and Yashunin, 2020) finds approximate nearest neighbours by greedy best-first traversal from entry points in a proximity graph. Strong general-purpose encoders include MPNet-based sentence transformers (Song et al., 2020), which we use, and Contriever (Izacard et al., 2022). Biomedical encoders are pre-trained on domain text (Lee et al., 2020; Gu et al., 2021) or trained contrastively on PubMed search logs (Jin et al., 2023). Dense models degrade outside their training distribution, and BEIR shows large out-of-domain gaps, including on biomedical tasks (Thakur et al., 2021). Sparse lexical retrieval with BM25

(Robertson and Zaragoza, 2009) remains a strong and complementary baseline.

2.3 Geometry of Embedding Spaces and the Query–Document Gap

Contextual embeddings are *anisotropic*: rather than spreading over the hypersphere, they occupy a narrow cone, so random pairs have high cosine similarity (Ethayarajh, 2019). Contrastively trained two-tower models can additionally exhibit a *modality gap*: the two input types are embedded in separate regions of the shared space, and the gap is created by initialisation and preserved by the contrastive objective (Liang et al., 2022). Questions and documents differ in length, style and vocabulary, so a bi-encoder applied to question answering faces an analogous *query–document gap*. In that case the query may lie far from the dense region of documents that answer it. Work on hard negatives shows that documents close to the query in embedding space are often not relevant (Xiong et al., 2021), and topic-aware training improves robustness to topically diverse queries (Hofstätter et al., 2021). Our preliminary results (Chapter 5) measure this gap directly in a biomedical corpus.

2.4 Re-ranking and LLM Relevance Judgements

Cross-encoders read the query and the document jointly and re-rank a candidate list far more accurately than bi-encoders, at a much higher cost per document (Nogueira and Cho, 2019). LLMs can be used as zero-shot re-rankers, pointwise or listwise (Sun et al., 2023), and as relevance assessors whose labels agree closely with human judgements (Thomas et al., 2024; Faggioli et al., 2023). The broader LLM-as-a-judge literature documents both the reliability and the biases of LLM evaluators (Zheng et al., 2023). Two limitations matter here. First, cost: every judged document is an LLM call. Second, resolution: pointwise judges asked for a label on a small integer scale assign the same label to many documents. Zhuang et al. (2024) show that finer-grained relevance labels and expected-relevance scores computed from token probabilities improve zero-shot LLM rankers. This is directly relevant to the tie problem we observe.

2.5 Query Reformulation and Relevance Feedback

Relevance feedback moves the query representation towards documents marked relevant and away from documents marked non-relevant. The Rocchio formula (Rocchio, 1971) has exactly this positive and negative form. Pseudo-relevance feedback (PRF) assumes that the top-ranked documents are relevant. ANCE-PRF learns a new query

embedding from the query and its top-k dense-retrieval results (Yu et al., 2021). LLM-based reformulation generates text in the document's register instead: HyDE embeds an LLM-generated hypothetical answer document and retrieves its neighbours (Gao et al., 2023a), and Query2doc appends a generated pseudo-document to the query (Wang et al., 2023). These methods change the *anchor* of retrieval. Our setting differs in that true relevance feedback, from the judge, arrives one document at a time and has a cost. The question is therefore how to choose which document to obtain feedback on.

2.6 Adaptive RAG and Retrieval-Failure Prediction

Several systems decide adaptively whether and how to retrieve. FLARE retrieves when the generator's confidence drops (Jiang et al., 2023). Self-RAG trains the generator to emit reflection tokens that critique retrieval (Asai et al., 2024). Corrective RAG (CRAG) uses a light retrieval evaluator to detect poor retrieval and trigger corrective actions such as web search (Yan et al., 2024). Query performance prediction (QPP) predicts retrieval quality before or after retrieval from properties of the query and of the score distribution (Cronen-Townsend et al., 2002; Carmel and Yom-Tov, 2010). These ideas motivate our gated policies: the expensive search should run only on queries where the cheap retriever is likely to have failed.

2.7 Search with an Expensive Evaluation Function

Our problem is an instance of a classical pattern: optimising an expensive black-box function with the help of a cheap surrogate under a budget. In heuristic search, best-first search expands nodes in the order of an evaluation function f (Pearl, 1984). A dense retriever is best-first search with $f = \delta(q, \cdot)$, and the number of expansions until the goal is reached is the rank of the target. Bayesian optimisation chooses the next point to evaluate by trading off exploitation and exploration of a surrogate model (Shahriari et al., 2016). Active learning (Settles, 2009) and preference elicitation in recommender systems (Rubens et al., 2015) choose the next query to an expensive oracle (a user or an annotator) to maximise information. Semantle-style games and trilateration localise a hidden point from distance feedback to probe points. Our contribution is to bring this view to RAG, with the LLM judge as the oracle and the embedding geometry as the surrogate.

2.8 Evaluation and Statistical Methodology

Retrieval experiments are usually compared on the same set of queries, so paired significance tests and an explicit analysis of statistical power are needed to avoid both false discoveries and under-powered conclusions (Smucker et al., 2007; Sakai, 2016). This is particularly important here, because the queries on which the retriever fails, where improvements can occur, are a minority of each dataset. Candidate benchmarks include PubMedQA (Jin et al., 2019), BioASQ (Tsatsaronis et al., 2015; Nentidis et al., 2023), MS MARCO (Nguyen et al., 2016) and Natural Questions (Kwiatkowski et al., 2019). Multiple-choice benchmarks such as MedQA (Jin et al., 2021) and MMLU (Hendrycks et al., 2021) are relevant for end-to-end evaluation but have no document-level ground truth.

2.9 Summary and Research Gap

Existing work either improves the cheap signal (better encoders, query rewriting, PRF), applies the expensive signal to a fixed candidate list (cross-encoder and LLM reranking), or decides *whether* to retrieve (adaptive RAG). To our knowledge, little work treats the LLM judge as an *expensive oracle inside a sequential search over the embedding space*, in which each judgement updates where to look next and the number of judge calls is the explicit cost. Little work has also related the success of such a search to measurable geometric properties of the query–document space. This proposal addresses both.

3. Problem Definition

This chapter formalises the problem discussed in our supervision meetings. The formulation started from a general question: find an item that maximises an expensive fitness function f, given a cheap surrogate f', per-evaluation costs $C_f \gg C_{f'}$ and a budget B. It was later specialised to retrieval, where the input is a set of documents, some of which have already been scored by the LLM, and the output is *which document to check with the LLM next*.

3.1 Setting

Let $D = \{d_1, \ldots, d_n\}$ be a corpus and let ϕ map text to the unit hypersphere $\mathbb{S}^{m-1} \subset \mathbb{R}^m$ ($m = 768$ in our experiments). For two texts x, y we use the cosine distance

$$\delta(x, y) \;=\; 1 - \langle \phi(x), \phi(y) \rangle \in [0, 2]. \tag{3.1}$$

A query q has a ground-truth (target) document $GT(q) \in D$, the document annotated as the source of its answer. Two evaluation functions are available:

- **Surrogate** $f'(d) = \delta(q, d)$. It costs $C_{f'} \approx 0$ per document, because document embeddings are pre-computed and indexed.
- **Judge** $J(q, d) \in \{0.1, 0.2, \ldots, 1.0\}$, an LLM's relevance rating of d for q (an integer 1–10 scale, normalised). It costs $C_f \gg C_{f'}$ per document: an LLM call with its latency and price.

3.2 Sequential Search with a Judge

A *search policy* π interacts with the judge in rounds. After t judge calls it holds the history $H_t = \{(d_i, J(q, d_i))\}_{i=1}^{t}$ and selects the next document to judge:

$$d_{t+1} \;=\; \pi(q, H_t) \in D \setminus \{d_1, \ldots, d_t\}. \tag{3.2}$$

The policy may use the query, the embeddings of all documents and the judged history, but no other information about the target. The search stops when the judge returns a terminal verdict (for example $J = 1.0$), when a stopping condition of the policy fires, or when the budget B is exhausted. It then returns $\hat{d} = \arg\max_{d \in H_t} J(q, d)$, with ties

broken by the surrogate.

We measure a policy with three complementary quantities:

1. **Cost to find the target:** $N_\pi(q)$, the number of judge calls until $GT(q)$ is judged. This is the "number of best-first-search iterations until the GT is found" used in our meetings. We report its mean and median over queries.
2. **Success under budget:** $\text{Succ@}B = \Pr_q[N_\pi(q) \le B]$, and the full *cost–accuracy curve* for $B \in \{1, 2, 3, 5, 10, 20\}$.
3. **Accuracy of the returned document:** whether $\hat{d} = GT(q)$ under a realistic stopping rule. A "soft" variant also accepts any document that the judge rates 1.0, because several documents can legitimately answer the same question.

Two feedback regimes. Because many documents can receive $J = 1.0$, a top score does not by itself identify the target. We therefore evaluate policies in two regimes. In the *verified* regime, the policy also learns whether a judged document is the target. This is the analogue of Semantle, where the player knows when the secret word has been hit, and it idealises a strong final verifier. It isolates *navigation* efficiency and yields $N_\pi(q)$. In the *judge-only* regime, the policy sees only J and must decide when to stop. This measures the accuracy of the returned document, and it exposes the cost of judge ties.

3.3 Static Policies as Best-First Search

Every fixed evaluation function $f : D \to \mathbb{R}$ induces a static policy π_f that judges documents in increasing order of f. For such a policy, $N_{\pi_f}(q)$ equals the rank of $GT(q)$ under f. The standard RAG retriever is the static policy with $f = \delta(q, \cdot)$, which we call **Baseline 1 (B1)**. Adaptive policies change f as feedback arrives. For example, they may move the search anchor towards the documents rated highly so far, away from documents rated low, or towards a location estimated by trilateration (Chapter 6).

3.4 Oracle Quantities Used Only for Analysis

Some geometric objects in our analysis require judging *every* document in the pool. The most important is the *oracle centroid*

$$C^\star(q) \;=\; \frac{\sum_{d\,:\,J(q,d)=1.0} \phi(d)}{\left\| \sum_{d\,:\,J(q,d)=1.0} \phi(d) \right\|}, \tag{3.3}$$

the normalised centre of mass of the documents the judge rates as fully relevant. Computing $C^\star$ costs $n \cdot C_f$, far beyond any realistic budget. We therefore use it only to diagnose the geometry and to bound what a policy could achieve. A deployable policy can only build an *online* estimate C_t from H_t.

3.5 Offline Evaluation

Because $J(q, d)$ has been pre-computed for every query–document pair in the pool, any policy can be replayed exactly and deterministically offline: each "judge call" becomes a table lookup. This makes it possible to compare many policies and hyper-parameters on identical inputs, at no additional LLM cost, and to reserve live LLM calls for the final end-to-end validation.

4. Research Objectives and Expected Significance

4.1 Main Objective

To develop and evaluate geometry-aware, judge-guided search policies that reduce the number of expensive LLM judge calls needed to retrieve the target document in a RAG system, focusing on the queries on which standard dense retrieval fails, and to explain their behaviour through the geometry of the query–document embedding space.

4.2 Research Questions

RQ1. The judge as a signal. To what extent is an LLM relevance judgement a reliable proxy for proximity to the target document in embedding space, and how can its resolution be increased so that ties among top-scored documents are broken?

RQ2. Geometry of failure. Which geometric properties of the query–document space explain dense-retrieval failures, and what distinguishes the queries on which the retriever fails from those on which it succeeds?

RQ3. Search policies. Can a policy that exploits judge feedback and embedding geometry find the target with fewer judge calls than the retriever's own order (B1), particularly on the queries where B1 fails?

RQ4. When to search. Can retrieval failure be detected early, for example from the judge's verdict on the first retrieved document or from the retriever's confidence, so that the costly search is invoked only when it is needed?

RQ5. Generality. Do the findings hold across embedding models, judges, and biomedical and general-domain datasets, and do retrieval gains translate into better end-to-end RAG answers?

4.3 Working Hypotheses

Table 4.1 lists the working hypotheses. Some were formulated and tested during the preliminary phase; the others will be tested in the proposed research.

Table 4.1: Working hypotheses and their current status.

	Hypothesis	Status
H1	Query embeddings lie further from document embeddings than documents lie from each other, so the neighbourhood of a query is sparse while the neighbourhood of its target is not.	Supported (§5.3)
H2	It is best to search mostly by distance to the query, but also a little by distance to the centroid of relevant documents.	Partially supported: +1 pp at most (§5.4)
H3	Judge scores decrease monotonically with distance to the target, so they provide a usable navigation signal.	Supported, with heavy ties (§5.2)
H4	Steering *towards* the centroid of judge-approved documents leads the search into a dense cluster of relevant-but-non-target documents. A *repulsive* (negative) weight on that centroid, as in Rocchio's negative-feedback term, reduces the cost on failed queries.	Open: naive attraction increases cost (§5.5)
H5	Invoking the adaptive search only when the first retrieved document is judged not to be the target (gating) preserves B1's cost on easy queries and reduces it on hard ones.	Open
H6	Trilateration from calibrated judge scores localises the target better than any single anchor (query or centroid).	Open
H7	Finer-grained or comparative judging reduces ties at the top score and thereby lowers the cost of every judge-guided policy.	Open

4.4 Expected Contributions and Significance

1. **A formulation** of LLM-judged retrieval as budget-constrained sequential search, which connects RAG with heuristic search, Bayesian optimisation and active learning, together with an offline evaluation protocol based on a large pre-scored query–document pool.
2. **An empirical account of the query–document geometry** in biomedical dense retrieval: the query is an outlier, documents are concentrated around the target, and judge-approved documents surround the target off the query–target axis. The account explains why simple centroid heuristics fail.
3. **Judge-guided search policies** (gated, repulsive feedback, trilateration) that reduce the number of LLM calls needed to recover from retrieval failures, with explicit cost–accuracy trade-offs.
4. **A characterisation and predictor of retrieval failure**, usable as a trigger for any corrective RAG component.
5. **Practical guidelines and open-source tooling** for cost-efficient RAG: when to call the judge, how many calls to budget, and which geometric features predict difficulty.

Significance. As RAG is deployed in clinical, legal and enterprise settings, LLM calls become a dominant operating cost, and silent retrieval failures become a dominant source of error. A method that spends judge calls only where they help, and spends as few as possible, improves both. Scientifically, the work clarifies how the geometry of learned embedding spaces limits nearest-neighbour retrieval, and how an expensive but accurate oracle can compensate for these limits.

5. Preliminary Results

The preliminary phase built the infrastructure described below. It validated the LLM judge, measured the geometry of the query–document space, and evaluated a first family of static and adaptive search policies. The results are presented in the order in which they shaped the research.

5.1 Experimental Setup and Dataset Construction

Pipeline. We implemented a modular RAG pipeline (LlamaIndex with a Chroma vector store) that accepts a query either as text or directly as a vector, and returns neighbours of an arbitrary point in embedding space. This capability is required by every policy that moves the search anchor. Documents and queries are embedded with `all-mpnet-base-v2` (768 dimensions, cosine distance) (Reimers and Gurevych, 2019; Song et al., 2020).

A first, easy setting. On PubMedQA (Jin et al., 2019) with its own corpus of about 272K abstracts, the dense retriever ranked the ground-truth abstract first for 186 of 199 questions (93.5%). This is far above the lexical baselines reported on the same corpus, and it leaves little room for improvement. The same setting gave the first evidence of a query–document gap. The K nearest neighbours of a question and the K nearest neighbours of its target overlap only modestly: the mean Jaccard overlap is about 0.30 at $K = 5$, peaks at about 0.40 at $K = 20$, and falls to about 0.24 at $K = 100$.

A challenging pool. Following the supervisors' recommendation, we built a harder benchmark on which the retriever succeeds in roughly 60–70% of the questions. We embedded the full PubMed corpus distributed with MedRAG (Xiong et al., 2024) (≈23.9M snippets) on a cloud GPU instance. For each of 199 PubMedQA questions we retrieved the 100 nearest neighbours of the question and the 100 nearest neighbours of its target. We interleaved these lists across questions and kept the first 5,026 unique documents as a *shared pool*; adding missing targets brings the pool to 5,035 documents. Every question–document pair ($199 \times 5{,}026 = 1{,}000{,}174$ pairs) was scored by **Gemini 2.5 Flash-Lite** with a rubric-based prompt on an integer 1–10 scale, normalised

to 0.1–1.0. Scoring was distributed over a rate-limited LLM gateway with a resumable, staged queue. Geometric analyses use the 196 questions for which a target embedding is available. On this pool B1 ranks the target first for **70.4%** of the questions, so it fails on 58 of 196, which is inside the intended range.

5.2 The LLM Judge as a Proximity Signal (RQ1)

Validating the judge. Our first judging prompt produced poorly calibrated scores, and an early conclusion was that "the LLM is not good as a judge". A redesigned rubric with well-separated score definitions changed this. In a pilot, the judge gave the highest score to the target for 192 of 197 questions, and the five misses appeared to stem from questionable ground-truth labels.

Correlation with distance to the target. Table 5.1 summarises the full-pool run. Per query, the judge's score correlates negatively with the cosine distance between the document and the target. The correlation is negative for *every* query and statistically significant ($p < 0.05$) for every query under Spearman, Pearson and Kendall. As a detector of the target the judge is almost perfect (ROC-AUC 0.997). Average precision is low, however, because the target is one of many documents that receive the top score.

Table 5.1: Judge validity on the full pool (199 questions, 1,000,174 scored pairs). Correlations are per-query medians between $J(q, d)$ and $\delta(d, GT)$.

Metric	Value
Spearman ρ / Pearson r / Kendall τ_b (median)	-0.29 / -0.50 / -0.23
Queries with negative / significant correlation	100% / 100%
ROC-AUC for identifying the target	0.997
Average precision (random baseline)	0.055 (0.0002)
Targets that receive the top score (1.0)	186 of 187 scored
Documents with score 1.0 per query (median / mean)	18 / 17.1

Score quantisation. The judge effectively uses six of its ten levels (Table 5.2), and the mean distance to the target decreases monotonically across them. The score is therefore an ordinal "distance gauge", which is the property a trilateration-style policy needs. Its resolution at the top is poor, though. A median of 18 documents per query share the score 1.0 with the target (the set G_3 below). In an earlier staged run the effect of ties was stark: MRR was 0.98 when ties were resolved in the target's favour, 0.26 when they were averaged, and 0.04 when they were resolved against the target.

Table 5.2: Score levels used by the judge (all pairs of the 196 analysed queries). The levels 0.3, 0.5, 0.7 and 0.9 together account for $< 0.02\%$ of the pairs.

Judge score	1.0	0.8	0.6	0.4	0.2	0.1
Share of pairs (%)	0.34	0.26	0.93	6.16	23.5	68.8
Mean $\delta(d, GT)$	0.215	0.404	0.632	0.771	0.841	0.892

The judge rarely flags a retrieval failure. For 46 of the 58 questions on which B1 fails, the judge gives the wrong top-ranked document the score 1.0, so it is "relevant" but not the annotated target. The first document receives a lower score in only 12 failed questions and in 1 of the 138 successful ones. As a failure detector, "$J(d_1) < 1.0$" is therefore precise (12 of 13) but has low recall (12 of 58). Two consequences follow. Evaluation must include a soft metric that accepts any fully relevant document. And gating the search on the first verdict (H5) requires a finer-grained judge (H7) or additional failure signals (RQ4). A pilot also indicated that the retriever's own top-1 confidence is markedly lower on failed queries, which makes it a candidate signal.

5.3 Geometry of the Query–Document Space (RQ2)

Queries are outliers (H1). For every query we drew two balls of equal radius $\delta(q, GT)/2$, one centred at the query and one centred at the target. The query ball contained *no* pool document for any of the 196 queries. The target ball contained other documents for 23% of the queries (38% of the failed ones). Even though the target is, by construction, the document the retriever is aiming for, the query lies in an empty region and the target lies in a populated one.

Three groups. For each query we defined three document sets: G_1, the documents within $\delta(q, GT)$ of the query; G_3, the documents the judge scores 1.0; and G_2, the documents within $\delta(C^\star, GT)$ of the oracle centroid $C^\star$ of G_3. G_1 is small (median 1 document, mean 3.1). G_3 is large (median 18) and contains the target for 94.9% of the queries. G_2 lies in between (median 6). The centroid is **44% closer to the target than the query**: the mean $\delta(\cdot, GT)$ is 0.097 for $C^\star$ against 0.173 for q, and $C^\star$ is closer for 92.3% of the queries. The gain is larger exactly where it is needed: the mean improvement is 0.110 on failed queries against 0.061 on successful ones, and it correlates with $\delta(q, GT)$ ($r = 0.72$).

The density anomaly. Although the centroid ball is *smaller* than the query ball, it contains about three times more documents. Two control experiments located the

cause in the geometry of the document space and not in the centroid itself (Table 5.3). Random points at the same distance from the target capture almost nothing. Any real pool document closer to the target than the query captures as many documents as the centroid. In addition, using such a document as a "simulated stronger judge" does not shrink the candidate set. Documents are concentrated in a dense region around the target, and the query lies outside it. A deep dive into the ten most extreme queries identified two mechanisms. In one, the centroid is pulled *away* from the target by distant false positives, yet still lands in a dense region. In the other, a tight and coherent G_3 cluster surrounds the target.

Table 5.3: Ball occupancy around different anchors (radius = anchor's distance to the target, except for random points, which use half the query's distance).

Anchor	Mean radius	Docs in ball	Docs / radius
Query q (G_1)	0.173	3.1	15
Random point near the target	0.087	0.8	11
Pool doc. closer to the target than q	0.06–0.10	10.1	60
Oracle centroid $C^\star$ (G_2)	0.098	9.7	93

The q–$C^\star$–target triangle. On the unit hypersphere, the triangle formed by the query, the oracle centroid and the target is nearly isosceles with its apex at the query. The mean spherical angles are $\angle q \approx 45.7°$, $\angle C^\star \approx 70.8°$ and $\angle GT \approx 70.6°$, with $\delta(q, C^\star) \approx \delta(q, GT) \approx 0.17$ and $\delta(C^\star, GT) \approx 0.10$. Seen from the query, the centroid and the target lie in directions about 46° apart. Moving from q towards $C^\star$ brings the search closer to the *cluster* of relevant documents, but not specifically to the target, which sits off the q–$C^\star$ axis.

5.4 Static Policies: Best-First Search with Fixed Evaluation Functions

To test H2 we compared static policies, each ranking the 5,035 documents by a fixed function f, with the oracle centroid as an optimistic ingredient (Table 5.4). Three sanity checks passed. B1 reproduces the RAG retriever exactly. H1 converges to B1 as $\alpha \to 1$. H2 (query distance, with centroid distance as a tie-breaker) is identical to B1 for all 196 queries, because 64-bit cosine distances never tie.

Even with oracle information, mixing in the centroid barely helps. The best hybrid (H1, $\alpha = 0.93$) gains one percentage point at top-1 but changes the target's rank in only 16 queries (6 better, 10 worse). Pure centroid ranking (B2) and re-ranking a short

Table 5.4: Static policies on 196 queries. Rank of the target = number of judge calls until it is found. B2 and H1–H3 use the *oracle* centroid $C^\star$. H1: $\text{score}(d) = \alpha \,\text{sim}(q, d) + (1 - \alpha)\,\text{sim}(C^\star, d)$. H3: the top-$k$ by $\delta(q, \cdot)$, re-ranked by $\delta(C^\star, \cdot)$.

Policy	Mean rank	Top-1 (%)	Top-3 (%)	Top-10 (%)	Mean rank (failed)
B1: $\delta(q, \cdot)$ (standard RAG)	**3.29**	70.4	**83.2**	**91.8**	**8.72**
B2: $\delta(C^\star, \cdot)$	9.87	14.8	36.2	62.8	14.31
H1, $\alpha = 0.93$	3.35	**71.4**	82.7	90.8	8.93
H1, $\alpha = 0.85$	3.40	70.9	82.7	90.8	9.09
H1, $\alpha = 0.70$	3.71	67.9	79.6	89.8	9.86
H2: $\delta(q, \cdot)$, tie-break $\delta(C^\star, \cdot)$	3.29	70.4	83.2	91.8	8.72
H3, $k = 10$	5.05	17.9	43.9	91.8	–
H3, $k = 50$	9.55	14.8	36.2	63.3	–

list by centroid distance (H3) collapse, as the triangle geometry predicts. A per-query table and three hand-traced queries confirmed that the policies behave as specified.

5.5 First Adaptive Policies

We then removed the oracle and let policies learn from the judge online.

- **Judge re-ranking of the top-k.** If ties are broken arbitrarily, top-1 accuracy collapses (40.8% at $k = 5$, 32.1% at $k = 10$). If ties are broken by query distance, judging the top-5 returns the target for 75.4% of the 187 queries with a scored target, against 73.8% for B1 on the same queries: five calls buy 1.6 points. The reason is that the wrong first document is usually also rated 1.0.
- **Iterative pseudo-relevance feedback.** Judging the top-k, re-ranking the whole pool by the centroid of the documents rated 1.0, and repeating, places the target at mean rank 4.9 with 31% top-1 ($k = 5$, one round), and more rounds make it worse.
- **Adaptive sequential search.** The policy follows B1 until the first document rated 1.0 is found. It then selects the next document by $\alpha \,\text{sim}(q, d) + (1 - \alpha)\,\text{sim}(C_t, d)$, where C_t is the online centroid of the documents rated 1.0 so far and is updated after every call. Every $\alpha < 1$ *increases* the cost (Table 5.5).

These negative results are informative. Attraction towards judge-approved documents moves the search into the dense cluster of relevant-but-non-target documents (on failed queries, the $\alpha = 0.5$ policy judges a median of 3.5 other documents rated 1.0 before it reaches the target). This is consistent with the triangle geometry, in which the target

Table 5.5: Adaptive sequential search: mean (median) number of judge calls until the target is found.

α	1.0 (=B1)	0.9	0.7	0.5	0.3	0.1
All 196 queries	3.29 (1)	3.35 (1)	3.72 (1)	4.10 (1)	4.43 (1)	4.55 (1)
58 failed queries	8.72 (4)	8.93 (5)	10.19 (5)	11.47 (6)	12.59 (8.5)	13.00 (9)

lies off the q–centroid axis.

5.6 Summary of Preliminary Findings

1. The judge is a strong but coarse proximity signal. Ties at the top score, not ranking errors, are its main limitation, and it rarely flags the retriever's own mistakes.
2. Queries lie outside the dense document region that surrounds their target, and judge-approved documents concentrate around the target *off* the query–centroid axis.
3. Neither static mixing (even with oracle information) nor naive online attraction to the centroid beats B1. Any gain must come from policies that (i) are invoked only when B1 is likely to fail, (ii) push away from relevant-but-non-target documents rather than towards them, or (iii) localise the target from *graded* feedback, which requires a finer-grained judge.

These three directions match the action items of the most recent supervision meeting (12.10.2026): analyse the failed queries in depth, evaluate on them separately, add a negative weight on the centroid distance, and start the adaptive search only once the first document is known not to be the target. They form the core of the proposed work.

6. Research Methodology and Work Plan

The research is organised in six work packages (WPs). Table 6.1 maps them to the research questions and hypotheses of Chapter 4.

Table 6.1: Work packages, research questions and hypotheses.

WP	Content	RQs	Hypotheses
WP1	Benchmark extension, offline simulator, statistical protocol	all	–
WP2	Characterisation and prediction of retrieval failures	RQ2, RQ4	H1, H5
WP3	Finer-grained judging	RQ1	H3, H7
WP4	Judge-guided search policies	RQ3, RQ4	H2, H4–H6
WP5	Generalisation across embedders, judges, datasets	RQ5	H1–H7
WP6	End-to-end RAG evaluation	RQ5	–

6.1 WP1: Benchmark, Simulator and Statistical Protocol

Larger query sets. The current pool contains 58 failed queries, too few to detect moderate effects on that subset. We will extend the benchmark to the full expert-annotated PubMedQA set (1,000 questions), using the same pool-construction recipe, so that B1 succeeds on about 60–70% of the questions. A full query×pool cross product grows quadratically, so we will pre-score each question's neighbourhood (neighbours of the question and of its target, plus sampled negatives). Any other document that a policy requests will be scored on demand and cached, which keeps the offline replay exact.

Power analysis. Using the per-query variance of the call differences measured in the preliminary phase, we will determine the number of failed queries needed to detect a target effect size, for example a 20% reduction in mean calls on failed queries, at $\alpha = 0.05$ and power 0.8. This determines the final benchmark size (Sakai, 2016).

Metrics and tests. For every policy we report N_π (mean and median), Succ@B and the cost–accuracy curve for $B \in \{1, 2, 3, 5, 10, 20\}$, and the strict and soft accuracy of the returned document (Chapter 3). Results are reported separately on all queries and on the failed-query subset. Policies are compared per query with paired Wilcoxon

signed-rank or randomisation tests, with bootstrap 95% confidence intervals and Holm correction for multiple comparisons (Smucker et al., 2007). Hyper-parameters are tuned on a development split and reported on a held-out split.

6.2 WP2: Characterising and Predicting Retrieval Failure

Deep dive into failed queries. Extending the hand-traced examples of the preliminary phase, we will analyse all queries on which B1 is not perfect and ask what separates them from the successful ones. *Analysis features* may use the target and the full judge scores: $\delta(q, GT)$, the target's displacement from the q–$C^\star$ axis, local density around q and around the target, and $|G_1|, |G_2|, |G_3|$. *Deployable features* use only what is available at query time: the similarity margin and spread of the top-k list, local density around the top documents, the judge's (refined) verdict on the first document, agreement among the top documents, and question type.

Failure prediction. From the deployable features we will train simple, interpretable classifiers (logistic regression and gradient-boosted trees) to predict "B1's first document is not the target". We will compare them with classical query-performance predictors (Cronen-Townsend et al., 2002; Carmel and Yom-Tov, 2010) using ROC-AUC and precision at fixed recall. The predictor is the trigger of the gated policies in WP4.

6.3 WP3: Finer-Grained Judging

The preliminary results show that ties at the top score, not ranking errors, limit the judge (H7). We will compare: (i) a sharper rubric that differentiates "answers the question" from "on-topic"; (ii) a finer scale (0–100); (iii) expected relevance computed from the token probabilities of the label (Zhuang et al., 2024); (iv) pairwise or listwise comparison, applied only *within* the tied top group, which bounds the extra cost; and (v) a stronger judge, including an open-weights model served on our cluster. Each variant is measured by the size of the top-score tie group, ROC-AUC and Kendall's τ against distance to the target, its accuracy as a first-document failure detector (WP2), and its cost per call.

6.4 WP4: Judge-Guided Search Policies

All policies are evaluated in the offline simulator, in both feedback regimes, against the same baselines.

Baselines. B1 (dense retrieval) and BM25 (Robertson and Zaragoza, 2009); cross-encoder re-ranking of the top-k (Nogueira and Cho, 2019); LLM re-ranking of the top-k with ties broken by query distance (Sun et al., 2023); Rocchio-style and ANCE-style pseudo-relevance feedback (Rocchio, 1971; Yu et al., 2021); HyDE and Query2doc (Gao et al., 2023a; Wang et al., 2023), at one generation call each; and the oracle-centroid policies of Chapter 5 as optimistic references.

(a) Gated search (H5). Judge B1's first document. If it is accepted, by the refined judge of WP3 and/or the failure predictor of WP2, stop after a single call. Otherwise, switch to one of the adaptive policies below. On the roughly 70% of queries where B1 is right, gating preserves B1's cost, so the comparison effectively happens on the failed queries, as suggested in our last meeting.

(b) Repulsive feedback (H4). Generalise the adaptive rule to

$$s_t(d) \;=\; \alpha\,\mathrm{sim}(q,d) \;+\; \beta\,\mathrm{sim}(C_t^{+},d) \;-\; \gamma\,\mathrm{sim}(C_t^{-},d), \tag{6.1}$$

where C_t^{+} is the online centroid of judged documents rated highly but rejected as the target, and C_t^{-} is the centroid of documents rated low. The preliminary results correspond to $\beta > 0, \gamma = 0$. We will study $\beta < 0$, a negative weight that pushes the search away from the cluster of relevant-but-non-target documents and towards the off-axis target, and $\gamma > 0$, Rocchio's negative-feedback term. A diversity penalty against already-judged documents, in the spirit of maximal marginal relevance, will also be tested.

(c) Trilateration from graded feedback (H6). This is the original algorithmic idea of the project, made feasible by the judge validation of Chapter 5.

1. *Calibration.* Fit a monotone map g from judge score to expected distance to the target (Table 5.2) on a development split.
2. *Initial anchors.* Judge a few anchors chosen by farthest-point sampling within the query's neighbourhood: start from the nearest document, then repeatedly add the document farthest from those already selected.
3. *Localisation.* Estimate the target position either (A) as a weighted centroid $\hat{x}_t \propto \sum_i w(J_i)\,\phi(d_i)$, or (B) by metric least squares on the sphere,

$$\hat{x}_t \;=\; \arg\min_{x\in\mathbb{S}^{m-1}} \sum_{i\le t} \big(\delta(x,d_i) - g(J_i)\big)^2 \;+\; \lambda\,\delta(x,q). \tag{6.2}$$

Because there are far fewer anchors than dimensions, x is restricted to the subspace spanned by q and the judged documents, which makes the problem well posed.

4. *Step.* Retrieve the nearest unjudged neighbour of $\hat{x}_t$ from the vector database (querying by vector), judge it, and update.
5. *Stopping.* Stop when a score exceeds a threshold, when there has been no improvement for a fixed number of iterations, or when the budget is reached. Return the best-judged document.

Finer-grained judging (WP3) directly sharpens g, so WP3 and WP4 will be iterated together.

(d) Query-adaptive weighting. Since the benefit of the centroid varies strongly across queries (it correlates with $\delta(q, GT)$), we will test choosing (α, β, γ) per query from the WP2 features.

6.5 WP5: Generalisation

We will repeat the geometric analysis (query outlier test, three groups, triangle, density controls) and the policy comparison with (i) other embedders: a biomedical retriever (Jin et al., 2023), Contriever (Izacard et al., 2022) and a stronger commercial embedding model; (ii) other judges; and (iii) other datasets with document-level ground truth: BioASQ (Tsatsaronis et al., 2015) in the biomedical domain, and MS MARCO (Nguyen et al., 2016) or Natural Questions (Kwiatkowski et al., 2019) in the general domain. Pools will be built with the same recipe, so that retrieval is comparably challenging.

6.6 WP6: End-to-End RAG Evaluation

The best policies will be integrated into the modular RAG pipeline and evaluated on downstream answer accuracy, following the MIRAGE protocol (Xiong et al., 2024) (PubMedQA yes/no/maybe and, where applicable, multiple-choice medical QA), together with total LLM calls, tokens and latency per question. This tests whether gains in retrieval cost and accuracy translate into better and cheaper answers.

6.7 Resources, Risks and Mitigation

Resources. The existing codebase (pipeline, LLM gateway, staged scoring queue, offline simulator, analysis suite), the BGU compute cluster (SLURM) and on-demand cloud GPUs for embedding, and an LLM API budget. Offline replay keeps API usage concentrated in one-time scoring runs.

Table 6.2: Main risks and mitigations.

Risk	Mitigation
Judge ties cap every policy	WP3 (finer judging); soft metrics; verified regime isolates navigation from ties
Too few failed queries for significance	Benchmark extension and power analysis (WP1)
LLM API cost and rate limits	Offline replay, caching, on-demand scoring only; open-weights judge on the cluster
Noisy or non-unique ground truth	Soft metrics; manual audit of a sample of disagreements
Policies do not beat B1	Gating alone already avoids judge calls on easy queries; failure characterisation and predictor remain standalone contributions

7. Time Schedule

The proposed research spans four semesters, from October 2026 to September 2028, and builds on the preliminary phase (November 2024 to October 2026) reported in Chapter 5. Figure 7.1 shows the plan at quarter-semester resolution, and Table 7.1 lists the milestones.

Figure 7.1: Research timeline. Each column is half a semester. S1: Oct 2026 to Mar 2027; S2: Apr to Sep 2027; S3: Oct 2027 to Mar 2028; S4: Apr to Sep 2028.

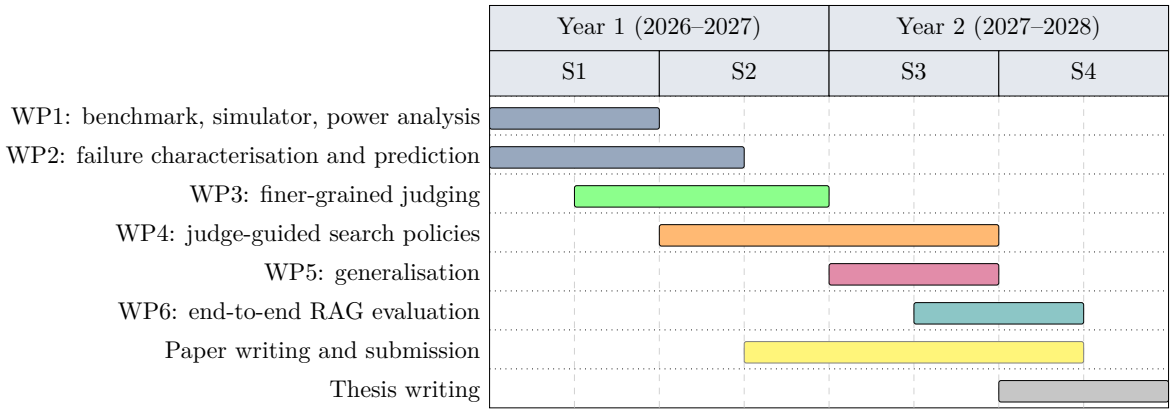


Table 7.1: Milestones.

	Date	Deliverable
M1	Mar 2027	Extended benchmark and power analysis; failure characterisation of all failed queries
M2	Sep 2027	Refined judge; gated and repulsive-feedback policies evaluated; first paper (judge signal and query–document geometry) submitted
M3	Mar 2028	Trilateration policy and cross-domain replication completed; main paper (judge-guided search) submitted
M4	Sep 2028	End-to-end evaluation completed; thesis submitted

Target venues are information retrieval conferences (SIGIR, ECIR, CIKM), NLP conferences (ACL, EMNLP and their Findings tracks), and, for the search-algorithmic contribution, the Symposium on Combinatorial Search (SoCS).

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תקציר

מערכות יצירה מועשרת-אחזור (Retrieval-Augmented Generation, RAG) מעגנות את תשובות מודל השפה (LLM) במסמכים שנשלפים ממאגר ידע חיצוני, ולכן איכות התשובה חסומה מלמעלה באיכות האחזור: אם המסמך הרלוונטי לא אוחזר, המודל אינו יכול להשתמש בו. עומדים לרשותנו שני סוגים של אות אחזור. מאחזרים צפופים מדרגים מסמכים לפי הדמיון בין הייצוג הווקטורי (embedding) שלהם לבין ייצוג השאילתה. הם זולים, אך בתחומים מתמחים הם מחטיאים את מסמך המטרה בחלק לא זניח מהשאילתות. שופטי LLM מעריכים רלוונטיות בדיוק רב בהרבה, אך כל קריאה אליהם יקרה, והציונים שלהם גסים: מסמכים רבים מקבלים את אותו ציון מרבי.

במחקר זה אנו מנסחים אחזור בעזרת שופט LLM כבעיית *חיפוש סדרתי תחת תקציב*. המרחק במרחב הייצוגים משמש פונקציית קירוב זולה, והשופט משמש אורקל יקר. בכל צעד מדיניות החיפוש בוחרת איזה מסמך לשלוח לשופט, במטרה להגיע למסמך המטרה במספר קטן ככל האפשר של קריאות. במחקר המקדים בדקנו 196 שאלות מ-PubMedQA מול מאגר משותף של 5,035 מסמכים שנשלפו מאינדקס צפוף של PubMed המלא, וכל כמיליון זוגות השאלה–מסמך דורגו על ידי Gemini 2.5 Flash-Lite. המחקר המקדים ביסס שלוש עובדות. (א) ציון השופט הוא קירוב אמין אך גס לקרבה למסמך המטרה: המתאם עם המרחק שלילי בכל השאילתות ו-ROC-AUC שווה ל-0.997, אך בחציון 18 מסמכים לשאילתה חולקים את הציון המרבי. (ב) השאילתות הן נקודות חריגות גיאומטרית ביחס למסמכים. (ג) מרכז הכובד (צנטרואיד) של המסמכים שהשופט אישר קרוב למסמך המטרה ב-44% יותר מהשאילתה, אך שימוש סטטי בו משפר את דיוק המקום הראשון בנקודת אחוז אחת לכל היותר (מ-70.4% ל-71.4%), והטיה נאיבית ומקוונת של החיפוש לכיוונו מייקרת את החיפוש.

על בסיס ממצאים אלה נפתח ונבחן: (א) מדיניות חיפוש מונחית שופט, שמופעלת רק כאשר סביר שהמאחזר נכשל, משתמשת במשוב דוחה ממסמכים רלוונטיים שאינם המטרה, וממקמת את המטרה בשיטת טרילטרציה מציוני השופט; (ב) שיפוט ברזולוציה גבוהה יותר, שמפרק שוויונות בציון; (ג) אפיון וחיזוי מוקדם של כישלונות אחזור. המדיניות תושווה באמצעות עקומות עלות–דיוק לאחזור צפוף, לדירוג מחדש בעזרת cross-encoder ובעזרת LLM, למשוב רלוונטיות מדומה ול-HyDE, על מאגרים ביו-רפואיים ועל מאגרים מתחום כללי.

מילות מפתח: יצירה מועשרת-אחזור, אחזור צפוף, גיאומטריה של מרחב הייצוגים, LLM כשופט, חיפוש תחת תקציב, מענה על שאלות ביו-רפואיות.

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חיפוש מונחה שופט LLM תחת תקציב קריאות

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